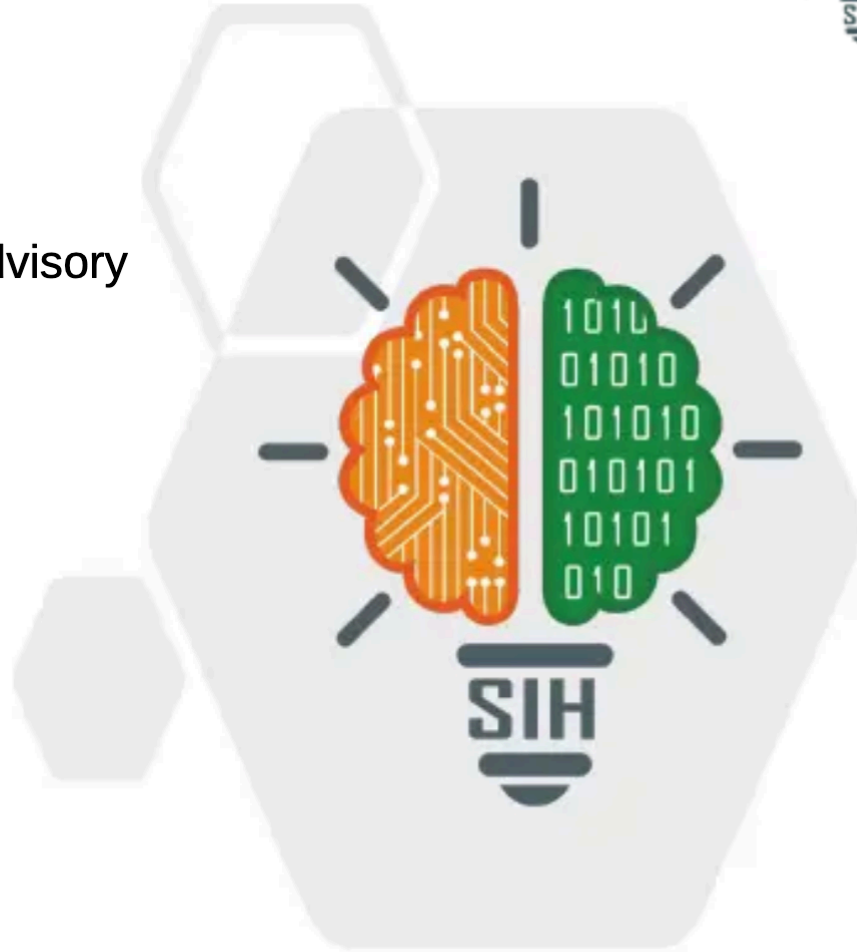


SMART INDIA HACKATHON 2025



SMART INDIA
HACKATHON
2025

- **Problem Statement ID** –25010
- **Problem Statement Title**- A Smart crop advisory system for small and marginal farmers
- **Theme**- Agriculture, Food Tech & Rural Development
- **PS Category**- Software
- **Team ID**- 49974
- **Team Name**- The Anomaly



KRISHI UNNATI

SMART CROP ADVISORY APP



Idea Resolution

AI-powered mobile app for Punjab's small farmers

- **Crop Disease Detection:** Identifies **crop diseases with AI** and provides timely **solutions**.
- **AI-driven Crop & Fertilizer Suggestions:** Offers **personalized crop and fertilizer recommendations** based on soil, weather, and crop history.
- **Local Language Voice Support:** Provides **voice-enabled assistance in multiple local languages**, designed for low-literacy farmers.
- **Pest/Disease Detection via Image Recognition:** **Detects crop pests and diseases** from uploaded images and suggests quick remedies.
- **Market Price & Weather Forecast Alerts:** Gives **real-time market price updates** and **accurate weather forecasts** for better planning and profit.
- **Offline Access for Low Internet Areas:** Allows access to saved data **without internet connectivity**. Works on **2G and auto-updates when online**.

Innovation & Uniqueness

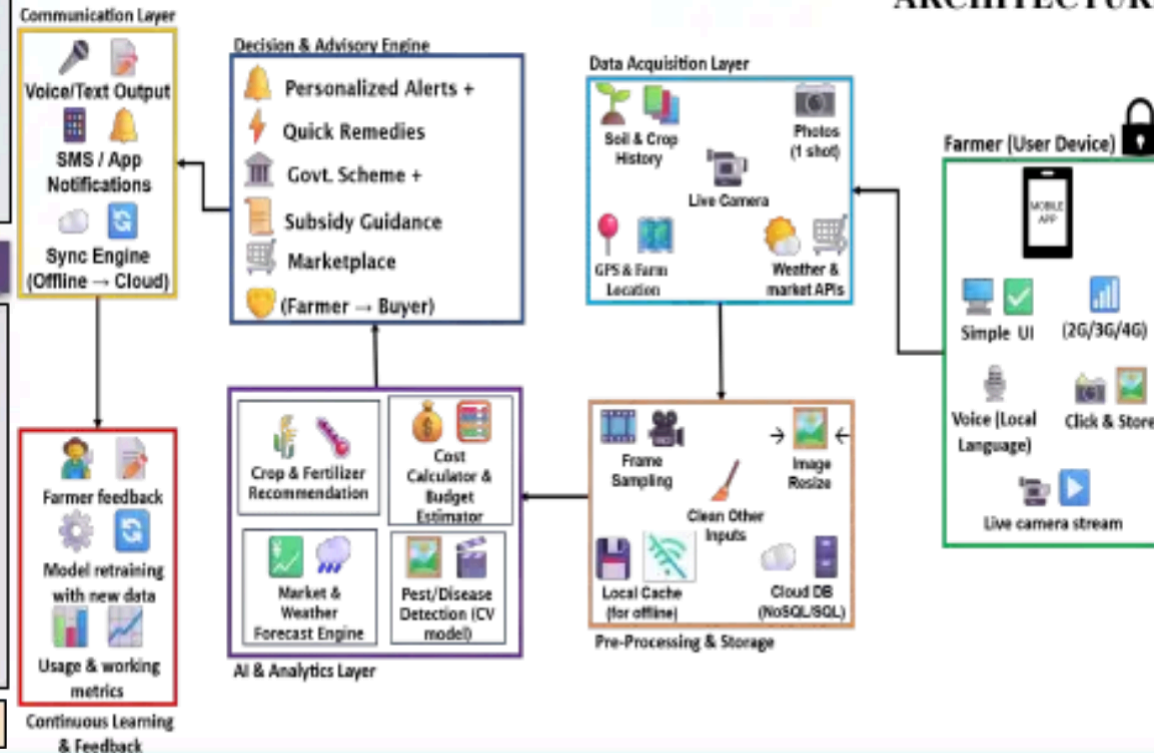
- **Marketplace** – Connect directly with buyers, track demand & prices, and earn higher profits without middlemen.
- **Farm Equipment** – Buy or rent tools & machinery, with all transactions auto-logged in the Farm Journal.
- **Advanced Intelligence** – Combines live camera pest detection, AI forecasts, and satellite insights for smarter farming decisions.
- **Pest Alert Map** – A community-driven early warning system for pest outbreaks.
- **Farm Journal** – Automatically logs daily farm activities for better planning & management – a true “Digital Twin of the Farm.”
- **Cost Calculator** – Estimates total input costs to support smarter budget planning.
- **Government Schemes** – Provides guidance on subsidies, financial aid, and support programs.

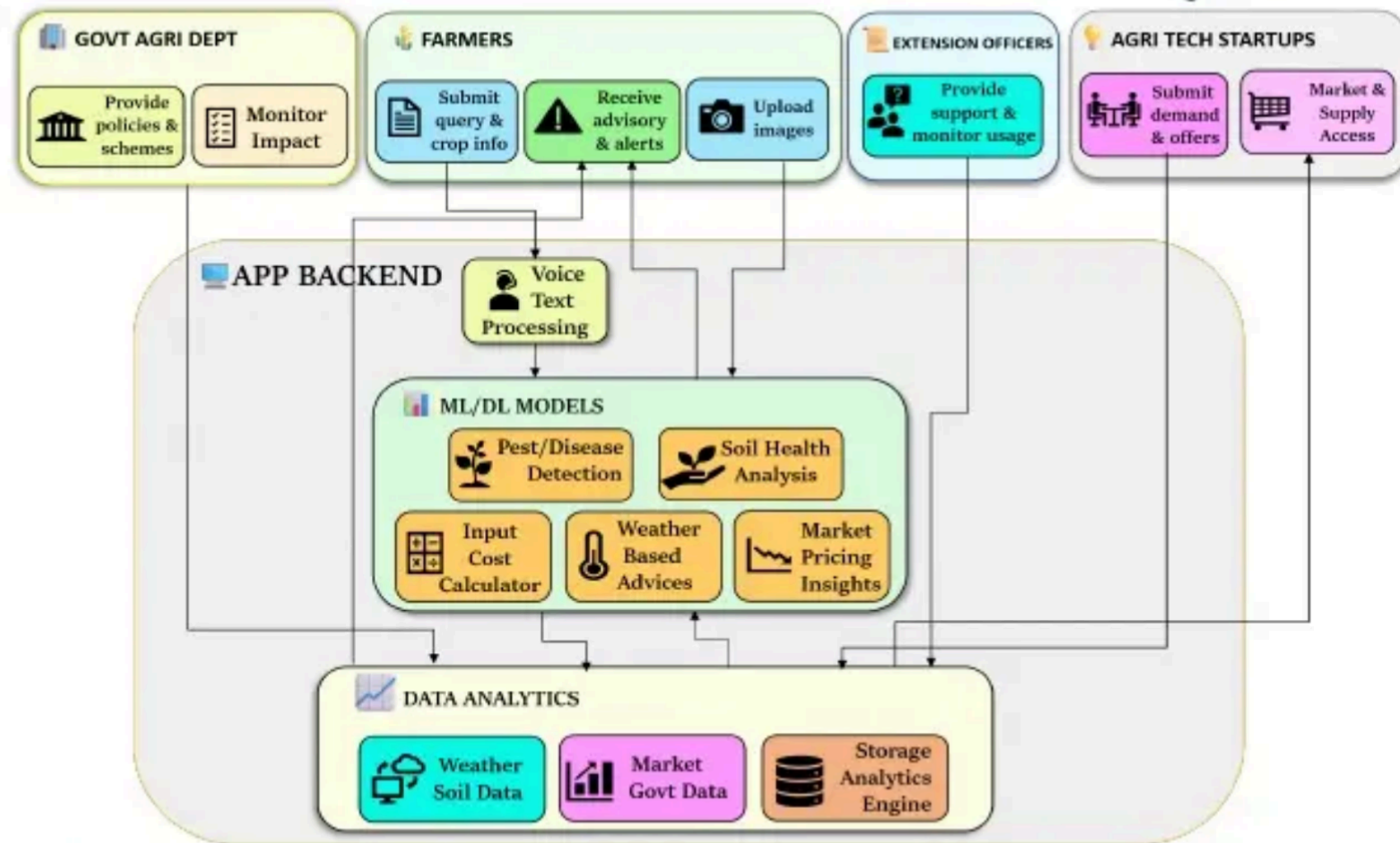
For a comprehensive analysis of our app features, please refer to this report: [\[Click Here\]](#)

Solution Approach

- Bridges info gap for **86%** of small farmers relying on guesswork
- Reduces input costs by **15-25%** through precision recommendations
- Increases yields by **20-30%** with predictive insights and alerts
- Ensures access on 2G networks; syncs when online

ARCHITECTURE





Project Status: Research ongoing, the prototype under development, and ML models built at a base level.

Challenges & Risks:

- **User Adoption:** Farmers may be **hesitant to trust or adapt** to new technology.
- **Low-End Devices:** Many users rely on old phones with **limited processing power and poor internet connectivity**.
- **Data Accuracy:** **Clean and reliable input data** is critical to maintain trust in recommendations.
- **Monetization:** **Balancing financial sustainability** while competing with free alternatives.

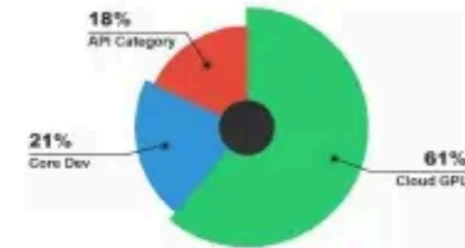
Feasibility:

- **All-in-One Support:** **Integrates disease detection, yield prediction, and buyer connections** — all in a single app.
- **Easy Access:** **Voice assistance and local language support** make it simple to use, even for low-literacy farmers.
- **Offline-First:** **Optimized for low-end 2G devices**, with automatic syncing when back online.
- **Scalable Model:** **Easily expandable to cover new crops and pests** as datasets grow.

Strategies:

- **Trust & Adoption:** Build credibility by **partnering with community leaders** and cooperatives, and **provide clear guidance in local languages**.
- **Data Integrity:** Ensure reliability through hybrid **AI combined with local knowledge**, and prioritize offline-first support for low-connectivity areas.
- **Financial Sustainability:** **Implement phased monetization** (marketplace fees, startup partnerships, promotional services), supplemented by government support for long-term viability.

Budget Breakdown:



Revenue Breakdown:



This revenue plan is a government recommendation; subsidies or additional revenue will depend on government policy decisions.

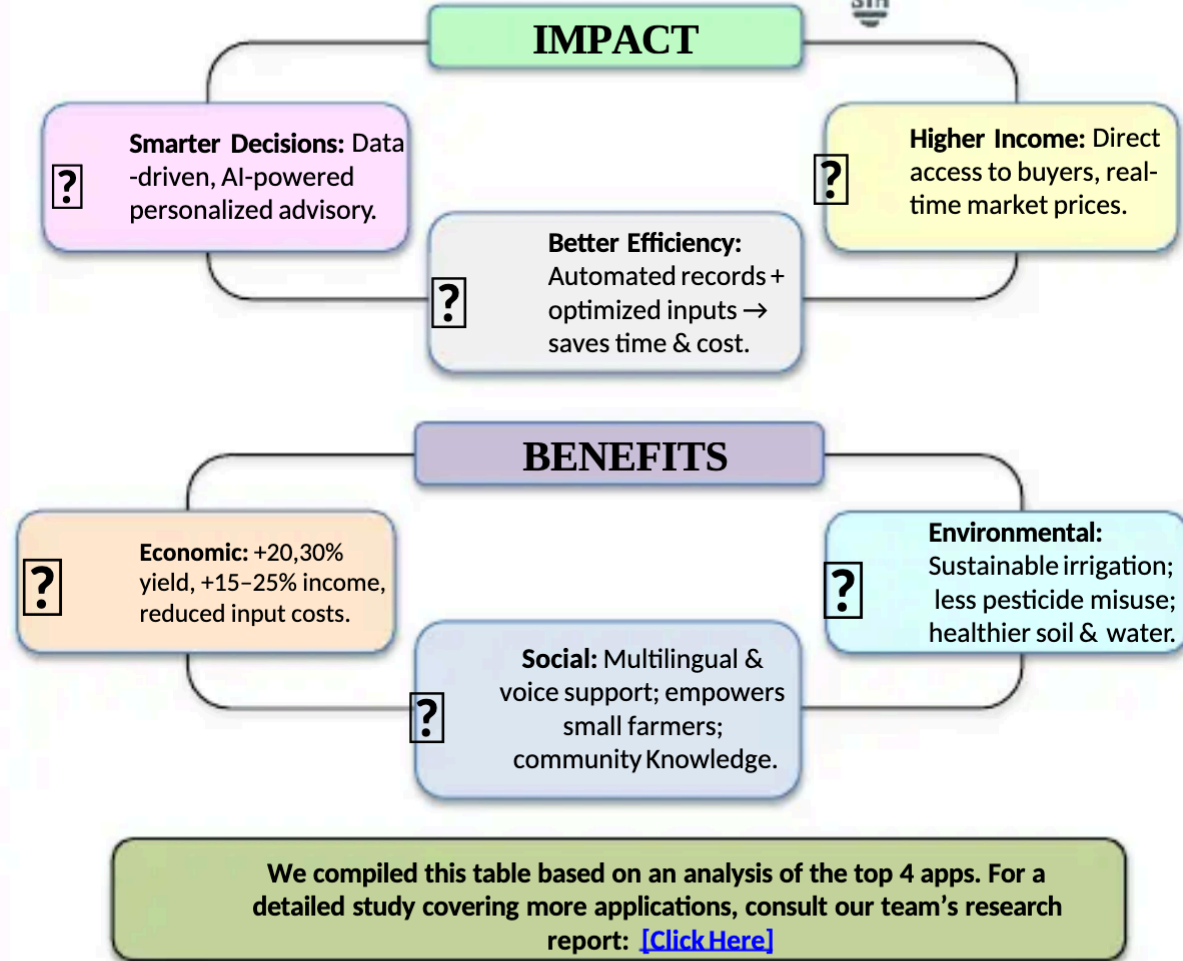
CORE DEV(INR MONTHLY)	
Database (MongoDB)	₹750-770
Google Cloud e2-micro	₹595-775
API CATEGORY(INR MONTHLY)	
SMS (Pfone)	₹155 for 1000 SMS
Maps (Google Maps)	₹15 for 1000 calls
Translation (Google Cloud)	₹1500 for 1M characters
CLOUD GPU(INR MONTHLY)	
Pest/Disease Detection(3 MODELS)	₹2000-5000
Soil Health Analysis	₹1000-2000
TOTAL(INR MONTHLY)	₹8100-8200

This is the most cost-effective budget for a small-scale prototype, just ₹8-9k/month. At scale, costs can be supported by Govt, and with features like marketplace commission, the model becomes self-sustainable. Additional costs may apply in the future.

For a detailed explanation of the budget breakdown and revenue sources, [\[click here\]](#).

Farming App Feature Comparison

FEATURES	Krishi Unnati ☑	Agri AI ☑	BigHaat ☑	Plantix ☑	Nurture.farm ☑
AI-Powered Crop & Fertilizer Suggestions	✓	✓	✓	✓	✗
Multilingual	✓	✗	✓	✓	✓
Voice Support	✓	✓	✓	✓	✗
Pest/Disease Detection via Image Recognition	✓	✓	✓	✓	✓
Market Price	✓	✗	✓	✗	✗
Weather Forecast Alerts	✓	✓	✓	✓	✓
Offline Access	✓	✗	✗	✗	✗
Farmer-to-Buyer Marketplace	✓	✗	✗	✗	✗
Farm Equipment Access	✓	✗	✓	✓	✓
Live Camera Pest Detection	✓	✗	✗	✗	✗
Pest Alert Map	✓	✗	✗	✗	✗
Farm Journal	✓	✗	✗	✗	✗
Cost Calculator	✓	✗	✗	✗	✗
Government Schemes	✓	✗	✗	✗	✗
AI-based Soil Health Analysis	✓	✓	✗	✗	✗



Research Articles:

- [An AI-Driven Smart Crop Recommendation and Advisory Framework](#)
- [Precision Agriculture and Smart Farming](#)
- [Exploring the Level of Digital Literacy](#)
- [Impact of Digitalization on the Farmers in India: Evidence using Panel Data Analysis](#)

Datasets & APIs:

- [Punjab Soil Health Maps \(PRSC\)](#)
- [Crop and Soil Dataset \(Kaggle\)](#)
- [India Government Commodity Prices API](#)
- [OpenWeatherMap API](#)

Reference:

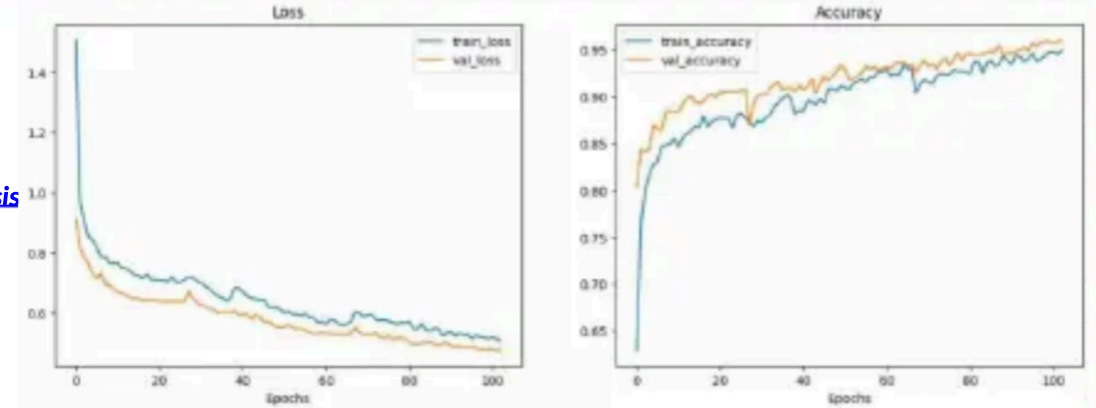
- [V. S. Sardeshmukh, S. B. Patil and Prof. S. Bedage, "An AI-Driven Smart Crop Recommendation and Advisory Framework", Journal of International Research Journal on Advanced Engineering Hub \(IRJAEH\). Vol. 3, pp \(3209-3218\), July 2025.](#)

Prototypes:

- <https://smart-crop-advisory-system.vercel.app/>
- [Pest Detection Model](#)

Supporting Reports & Findings:

- [Krishi Unnati Research Reports](#)
- [Features of Krishi Unnati](#)
- [Comparison with Other Agri Apps](#)
- [Budget & Revenue Breakdown](#)
- [Machine Learning Model Development & Evaluation](#)



These Supporting Reports & Findings are our self-prepared extended documents, created from our research and observations. They provide a detailed review and deeper understanding of our app.